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A study of lights at night suggests dictators lie about economic growth

Satellite data hints at the scale of their deception

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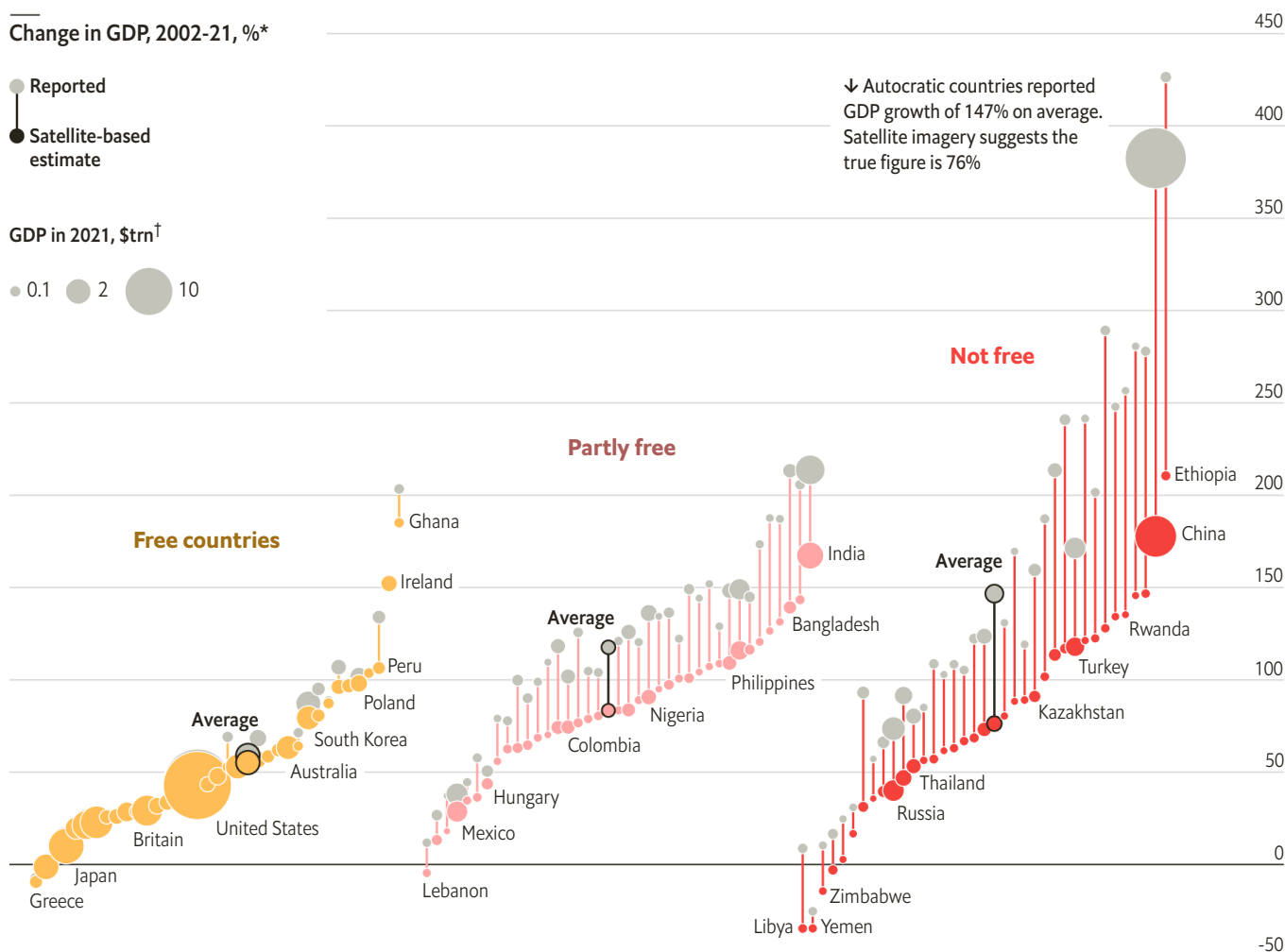
Sep 29th 2022 | 3 min read

Change in GDP, 2002-21, %*

● Reported
● Satellite-based estimate

GDP in 2021, \$trn[†]

● 0.1 ● 2 ● 10



*Countries with over 5m people, freedom status in 2021 [†]In 2021 \$ at market exchange rates, assuming reported 1992 GDP figures are accurate

BENITO MUSSOLINI was a tyrant, but at least he made the trains run on time. Or so the story goes. Dictators are often seen as ruthless but effective. Official GDP figures support this view. Since 2002 average reported economic growth in autocracies has been twice as fast as in democracies.

But in fact, Mussolini's trains often ran late—and dictators' economic stewardship may not be as effective as they claim. New research finds that autocrats greatly overstate their countries' economic growth.

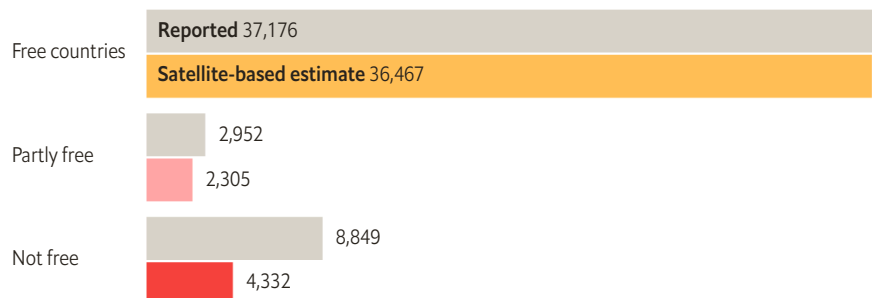
In a peer-reviewed article that will be published this month, Luis Martinez, an economist, investigated dictators' GDP-growth figures. To do so, he first obtained data on the brightness of countries' lights at night, as measured by satellites, a well-known proxy for GDP. He combined it with data from Freedom House, a think-tank, on countries' political systems. Assuming that the most democratic countries reported growth figures accurately, he then used the satellite data to estimate if other countries under- or over-stated theirs.

The data showed that dictators' reported GDP tended to grow much faster than satellite images of their countries would suggest. This could not be explained by their economies being based on different industries from other countries, or that people

economies being based on different industries from other countries, or that people there had lower average incomes.

Curious patterns in the data suggest manipulation as the cause. Mr Martinez found that the mismatch between satellite and GDP data did not appear in dictatorships until they were too rich to receive some types of aid: only showing up when governments would not forfeit money. The irregularities were most prevalent in the parts of GDP figures that are easiest to manipulate such as investment and government spending, and was bigger when these countries' growth was low compared with others'. And as countries moved towards or away from dictatorship, their numbers grew more or less suspicious.

Average GDP per person, 2021, \$*



*In 2021 \$ at market exchange rates, assuming reported 1992 GDP figures are accurate

The differences between reported and estimated GDP-growth rates were large. While others have found similar relationships, Mr Martinez was able, using satellite data up to 2013, to estimate the bias more precisely. In updated figures he has provided to us, cumulative GDP growth between 2002 and 2021 in countries "not free" is nearly cut in half: from 147% to 76%.

The explanation is probably simple: opportunity and motive. Part of what makes dictatorships dictatorships is that questioning the official line is dangerous. At the same time, autocratic regimes have a strong incentive to report healthy growth: its absence may be taken as a sign of incompetence or weakness, which dictators can ill afford.

Autocrats' subordinates face similar incentives. In a related study Jeremy Wallace, a researcher, found misreporting by Chinese provinces, too. As he notes, a leaked American diplomatic cable from 2007 revealed the view of Li Keqiang, the prime minister, then a provincial party secretary. He had said, with a smile, that GDP figures were "for reference only": he relied instead on proxies, such as electricity use.

Like their leaders, citizens in dictatorships often assume they are being lied to. Outsiders should be similarly sceptical. ■

Chart sources: "How much should we trust the dictator's GDP growth estimates?", by L.R. Martinez, 2022; Freedom House; World Bank

This article appeared in the Graphic detail section of the print edition under the headline "Shining light on lies"

60%

70%

→ **A study of lights at night suggests dictators lie about economic growth**



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